Technology Stack

Project: Employee Software Installation Request (ServiceNow)

1. Platform

The project is developed using the **ServiceNow Personal Developer Instance (PDI)**, which provides a cloud-based environment for designing, developing, and testing ServiceNow applications.

Component	Details
Platform	ServiceNow Personal Developer Instance (PDI)
Application Scope	Custom Scoped Application – *Software Installation Request*
Hosting Environment	Cloud-hosted ServiceNow Developer Instance (developer.servicenow.com)

2. ServiceNow Modules Used

The following ServiceNow modules were used to implement the software installation request workflow.

Module	Purpose
Service Catalog	Provides a user-friendly request form where employees can submit software installation requests by entering software details, business justification, installation device, license type, and required installation date.
Flow Designer	Automates the complete workflow, including manager approval routing, approval and rejection handling, catalog task creation, and email notifications.
Business Rules	Executes server-side automation, such as automatically creating an Incident when a software request cannot be fulfilled because of license shortages.
Approval Engine	Manages manager approvals using the **sysapproval_approver** table and provides the **My Approvals** interface for reviewing requests.
Access Control Lists (ACLs)	Implements role-based security by controlling who can view, approve, or modify software requests.
ServiceNow Studio	Provides an integrated development environment for building and managing the scoped application, workflows, scripts, and other application components.

3. Programming and Workflow Logic

The project combines scripting with low-code workflow automation to implement business logic efficiently.

- **Programming Language:** JavaScript
- **Server-Side APIs:** GlideRecord, GlideDateTime, and other ServiceNow APIs

- **Workflow Automation:** ServiceNow Flow Designer
- **Automation Features:**
  - Manager approval routing
  - Automatic task creation
  - Email notifications
  - Incident generation for license shortages
  - Dynamic field mapping using Data Pills

4. Data Storage

The application stores and manages request information using standard ServiceNow tables.

Table Name	Purpose
sc_cat_item	Stores the Service Catalog item definition and associated variables.
sc_request	Stores the parent request created when an employee submits a software request.
sc_req_item	Stores the individual requested item and tracks its approval and fulfillment lifecycle.
sysapproval_approver	Stores manager approval records and approval status.
sc_task	Stores fulfillment tasks assigned to the Network/Software Support team after approval.
incident	Stores incidents automatically generated when software requests cannot be fulfilled due to license shortages.

5. Version Control and Documentation

Proper version control and documentation practices were followed throughout the project development.

Component	Purpose
GitHub	Used for version control, project backup, and source code management through Studio Source Control or Update Set export.
Project Documentation	Prepared using Microsoft Word and PDF formats for project reports, diagrams, and technical documentation.
README.md	Used to provide project overview, setup instructions, and implementation details.

6. Development Tools

The following tools were used during the design, development, testing, and maintenance of the project.

Tool	Purpose
ServiceNow Studio	Develops and manages the custom scoped application.
Flow Designer	Designs and automates the software request workflow.
Business Rule Editor	Implements server-side business logic and automation.
Git & GitHub	Maintains version control and project history.
Web Browser Developer Tools	Used for testing, debugging, and validating client-side functionality.
Postman (Optional)	Used for testing ServiceNow REST APIs when required.

Summary

The **Employee Software Installation Request** project is built entirely on the **ServiceNow platform**, utilizing native features such as the **Service Catalog**, **Flow Designer**, **Business Rules**, **Approval Engine**, and **Access Control Lists (ACLs)**. JavaScript is used for implementing server-side business logic, while ServiceNow's standard database tables manage requests, approvals, tasks, and incidents. GitHub is used for version control, and ServiceNow Studio provides a centralized environment for application development. This technology stack enables the development of a secure, scalable, and maintainable workflow automation solution.